



AMITY UNIVERSITY MAHARASHTRA
AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B. Tech Computer Science & Engineering
ACADEMIC YEAR: 2025-26(ODD)

Course	:	B.Tech (CSE)
Course Code & Subject:		CSE 2301 & DCCN
Semester & Year	:	5 th Sem / 3 rd year
Division/Batch	:	A

ASSIGNMENT-03

1. Explain in detail the layered architecture of computer networks. Discuss how encapsulation and de-capsulation occur during data transmission.
2. Discuss error detection and correction techniques such as CRC and Hamming Code with examples.
3. Explain multiple access techniques such as CSMA/CD, and CSMA/CA & Compare their efficiency.
4. Describe the IPv4 addressing scheme and explain the process of sub-netting with an example.
5. Explain error detection and correction mechanisms using Checksum and Hamming Code. Compare their efficiency.
6. Explain the implementation and working of Network Address Translation (NAT). Discuss its advantages and limitations.
7. What are the IEEE 802.3 and IEEE 802.11 standards? Explain briefly.
8. Explain the concept of transmission impairments and data rate performance.
9. Describe the sliding window protocol used for flow control.
10. State the differences between IPv4 and IPv6 header formats.

Instructions:

1. **Attempt any 6 questions.**
2. Neatly write answers with proper diagrams wherever necessary.(consider mark weightage for each question: 10M)
3. Late submission will not be entertained.
6. Attach a cover page mentioning your Name, Roll Number, and Division.

Submission Deadline: on or before 21st Nov 2025.